PREDICTASTOCK AI TERMINAL

HOW STOCKS ARE PREDICTED: FACTORS, MATHEMATICS & AI EXPLAINED

Methodology Report * Target Markets: NSE / BSE * Date: 12 October 2026 * Inference: Groq (gpt-oss-120b)

1. EXECUTIVE SUMMARY: IS IT MATH, AI, OR BOTH?

The Core Architectural Distinction Between Mathematical Boundaries and AI Reasoning

A common misconception in retail trading software is that an AI Large Language Model (LLM) 'predicts' a future stock price. In institutional finance, allowing an LLM to guess future prices is dangerous because LLMs suffer from ungrounded hallucinations and cannot calculate physical volatility bounds.

PredictaStock uses a HYBRID QUANTITATIVE-AI ARCHITECTURE with strict division of responsibility:
1. The MATHEMATICS (100% Deterministic) calculates the exact price targets, upper resistance envelopes, and stop-loss boundaries using verified 14-period Average True Range (ATR-14) formulas.
2. The QUANTITATIVE MULTI-FACTOR ENGINE screens and ranks all 60+ Nifty constituents across 4 weighted pillars to find the 5 highest-probability opportunities.
3. The ARTIFICIAL INTELLIGENCE (Groq Cloud running openai/gpt-oss-120b) acts as an Adversarial Committee of 7 Specialist Analyst Desks to debate qualitative catalysts, identify tailwinds, challenge bull vs. bear theses, and calibrate the final trade execution plan.

2. THE 4 QUANTITATIVE PILLARS THAT DECIDE THE 5 STOCKS

Mathematical Weighting Model Generating the Composite Score (0 to 100)

Every single stock in the Nifty 50 and Nifty 100 universe is ingested and scored across 4 foundational pillars using the formula:
Composite Score = (0.35 * Technical) + (0.25 * Fundamental) + (0.25 * News Sentiment) + (0.15 * Macro Flow)

Quantitative Pillar	Weight	Specific Factors & Mathematical Inputs Evaluated
Pillar 1: Technical Momentum	35% Weight	RSI-14 (sweet spot 45 to 68), price confirmed above 20-DMA trendline, positive MACD histogram exp
Pillar 2: Fundamental Durability	25% Weight	Tier-1 Nifty 50 core leadership (75 pts baseline), balance sheet solvency, conservative debt-to-equity
Pillar 3: News & Sentiment	25% Weight	Live rolling 7-day Google News RSS scraping with financial keyword tone analysis. Normalizes score
Pillar 4: Macro & Liquidity Flow	15% Weight	Benchmark Nifty 50 directional drift, India VIX regime (<15 calm bullish vs >20 high volatility caution),

The News Veto Rule (Section 6.6):

Regardless of whether a stock has a perfect technical breakout or low valuation, if live news sentiment scores <= -0.40 (indicating severe regulatory orders, auditor resignations, fraud allegations, SEBI investigations, or sudden management turmoil), the stock is INSTANTLY DISQUALIFIED. The system refuses to recommend a stock facing existential headline risk.

3. HOW THE 5-DAY TARGET IS CALCULATED (THE MATH)

ATR-14 Volatility Grounding: Eliminating Hallucinations

The price target is NOT an opinion. It is derived mathematically from the stock's actual trading volatility over the last 14 sessions:

Formulas for the 5-Day Volatility Envelope:

Mathematical Metric	Formula Description
True Range (TR)	TR = max[(High_t - Low_t), |High_t - Close_t-1|, |Low_t - Close_t-1|]
Average True Range (ATR-14)	ATR_14 = (1 / 14) * SUM(TR_i) across 14 historical trading sessions
5-Day Expected Target Price	Target = CMP + (1.30 * ATR_14)
Upper Volatility Resistance Band	Range_High = CMP + (1.50 * ATR_14)
Lower Support / Invalidation (SL)	Range_Low = CMP - (1.20 * ATR_14)
Projected Gain Percentage	Expected Appreciation % = ((Target - CMP) / CMP) * 100

Real Worked Numerical Example (Vedanta Ltd - VEDL):

To understand how this operates in practice, consider the top pick currently displayed on your terminal:
- Stock: Vedanta Ltd (VEDL) on NSE
- Current Reference Price (CMP): Rs 264.90

- Measured 14-Day True Volatility (ATR-14): Rs 7.43
- 5-Day Target Calculation: Target = Rs 264.90 + (1.30 * 7.43) = Rs 274.57 (+3.65%)
- Upper Range High: Rs 264.90 + (1.50 * 7.43) = Rs 276.05 (Upper Volatility Band)
- Lower Invalidation (Stop Loss): Rs 264.90 - (1.20 * 7.43) = Rs 256.00 (Lower Support Band)

Notice that because VEDL's historical volatility is Rs 7.43, its expected 5-day gain is capped at +3.65%. The math prevents the system from making false promises like +20% on a stock that physically moves only Rs 7 a week.

4. THE ROLE OF THE AI: 7 SPECIALIST ANALYSTS & GROQ LLM

Multi-Agent Committee Debate Powered by Groq (openai/gpt-oss-120b)

Once the mathematical engine selects the top 5 stocks, the Artificial Intelligence is summoned. Connected to your active Groq Cloud key running the 120-Billion parameter 'openai/gpt-oss-120b' model, the AI simulates 7 specialized institutional desks:

Specialist Analyst Desk	Committee Stance	Institutional Focus & Responsibilities
1. Technical Analyst	BULLISH	Validates RSI-14 momentum, 20-DMA support, and volume breakout structure.
2. Fundamental Analyst	ACCUMULATE	Evaluates ROCE, balance sheet resilience, and dominant sector market share.
3. Options & Volatility Desk	FAVORABLE RANGE	Analyzes implied volatility smile and options strike support levels.
4. News & Catalysts Desk	POSITIVE MOMENTUM	Evaluates recent quarterly reports, contracts, and expansion timelines.
5. Sentiment & Media Desk	BULLISH	Measures social and financial media sentiment and regulatory disclosure health.
6. Sector Rotation Desk	OVERWEIGHT	Measures relative strength index against benchmark Nifty 50.
7. Risk Manager Desk	CALIBRATED RISK	Establishes position sizing (max 5-8% allocation) and downside invalidation.

The Adversarial Debate Arena:

The AI does not simply say 'buy'. It runs an adversarial debate:
- Bull Agent: Defends the 3 strongest upside thesis points.
- Bear Agent: Identifies the 3 primary risk factors and failure hurdles.
- Facilitator Consensus: Synthesizes a unified verdict (e.g. 'BUY / ACCUMULATE ON DIP').
- Calibrated Trade Plan: Delivers exact Suggested Entry Zone, Stop Loss (lower ATR band), and 5-Day Target.

5. STEP-BY-STEP PIPELINE: FROM UNIVERSE TO USER SCREEN

The 10-Stage Deterministic Workflow

Stage	Pipeline Phase	Automated Execution Actions
Stage 1	Universe Ingestion	Ingests 60+ constituents across Nifty 50 and Nifty 100 universe.
Stage 2	Data Verification	Pulls 6-month historical OHLCV data; verifies no 0.5% price discrepancies.
Stage 3	Indicator Computation	Calculates RSI-14, 20-DMA, ATR-14, and MACD divergence vectors.
Stage 4	Live News RSS Scrape	Scrapes rolling 7-day headlines via Google News RSS for each ticker.
Stage 5	Veto Rule Filtering	Disqualifies any scrip with sentiment <= -0.40 or severe regulatory actions.
Stage 6	Composite Scoring	Applies 4-pillar weights to compute score (0-100); filters for score >= 68.0.
Stage 7	Shortlist Selection	Selects the Top 5 unique, sector-diversified stocks with highest composite scores.
Stage 8	ATR Bounds Math	Calculates exact target (CMP + 1.30*ATR) and lower invalidation levels.
Stage 9	7 AI Committee Debate	Dispatches shortlist to Groq LLM to synthesize bull vs bear trade plans.
Stage 10	Immutable Logging	Saves records with immutable UUIDs to SQLite for 5th session alpha evaluation.

ACCOUNTABILITY PROMISE: BENCHMARKED AGAINST NIFTY 50

At the close of the 5th trading session, every forecast is compared against actual market closing prices and the Nifty 50 baseline return to compute real institutional Alpha. If the stock beats the index, it is graded a HIT; if it underperforms, it is recorded as a MISS. Past predictions cannot be edited, deleted, or revised.